SIMATS ENGINEERING
ASSESSMENT TOOL 2: Industry Problem Solving Task

COURSE NAME: COMPUTER VISION COURSE CODE: ITA0510 Slot A

COURSE OUTCOME COVERED
CO4: ***Compare object and pattern recognition techniques for interpreting visual data with spatial and motion characteristics. (BTL 5)*** **Assessment Tool Weightage: 40%**

QUESTIONS:5 Total Marks 50 Duration :60 Minutes Annexure A
Industry Problem Solving Task

Code of Conduct:

I P.JYOTHIKA(192311156) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Rubrics for Evaluation

Criteria	**Weight age (%)**	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**	***Marks(100)***
Understanding of Industry Problem	20%	Comprehensive understanding of real world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding; misinterprets problem context	
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts	
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution	
Use of Tools	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools	
Analysis	10%	Thorough analysis with validated results and	Good analysis with reasonable	Basic analysis with limited validation	Poor or no analysis	

		performance evaluation	validation			
Professionalism	5%	Highly professional, well documented, and clearly presented work	Good documentation and presentation	Basic documentation with some gaps	Poor documentation and presentation	

Analyze the given problem by comparing available techniques against the specified constraints and requirements. Evaluate and justify your decision using logical reasoning, clearly indicating how the chosen approach satisfies the conditions.

1. Smart Surveillance System Selection
A city surveillance system must operate under the following conditions:

- If latency > 100 ms → system fails
- If accuracy < 85% → unacceptable
- Hardware supports only moderate computation

Models available:
Viola-Jones: 78%, low latency
YOLO: 92%, moderate latency
Deep Learning: 95%, high latency
Compare the models and evaluate which satisfies all constraints. Justify your decision using logical reasoning based on given conditions.

ANSWER:

1. Smart Surveillance System Selection

Model	Accuracy	Latency	Computation	Result
Viola-Jones	78%	Low	Low	❌ Fails accuracy
YOLO	92%	Moderate	Moderate	✅ Satisfies all
Deep Learning	95%	High	High	❌ Fails latency/computation

Decision: YOLO

- Accuracy requirement is **≥85%**. YOLO gives **92%**, so it passes.
- Latency must be **≤100 ms**. YOLO has moderate latency, so it is suitable.
- Hardware supports only **moderate computation**, which matches YOLO.
- Viola-Jones is fast but its **78% accuracy is unacceptable**.
- Deep Learning has better accuracy, but its **high latency and computation** violate the

constraints.

Conclusion: YOLO is the best choice because it satisfies accuracy, latency, and hardware constraints simultaneously.

2. Autonomous Vehicle Detection Logic
A vehicle detection system must:
- Prioritize safety (accuracy > 90%)
- Maintain response time < 50 ms

Options:
Method A: 88% accuracy, 30 ms
Method B: 93% accuracy, 70 ms
Method C: 91% accuracy, 45 ms
Compare the methods and evaluate which should be selected. Justify your choice using constraint-based reasoning.

ANSWER:

Autonomous Vehicle Detection Logic

Method	Accuracy	Response Time	Result
Method A	88%	30 ms	❌ Accuracy too low
Method B	93%	70 ms	❌ Response too slow
Method C	91%	45 ms	✅ Satisfies both

Decision: Method C

- Safety requires accuracy >90%.
- Response time must be <50 ms.
- Method A is fast but has only 88% accuracy.
- Method B has good accuracy (93%) but takes 70 ms, which is too slow.
- Method C provides 91% accuracy and 45 ms response time, satisfying both constraints.

Conclusion: Method C should be selected because it meets both safety and response-time requirements.

3. Motion Analysis Decision Problem
A traffic system requires:
- Accurate motion direction detection
- Robustness in dense traffic

Results:

Optical Flow: accurate but noisy in dense scenes
Motion Estimation: stable but less detailed
Compare both approaches and evaluate which is suitable under high-density conditions. Justify using scenario-based logic.

ANSWER:

Motion Analysis Decision Problem

Approach	Direction Accuracy	Dense Traffic Robustness	Result
Optical Flow	Accurate	Noisy in dense scenes	❌ Less suitable
Motion Estimation	Less detailed	Stable	✅ More suitable

Decision: Motion Estimation

- The system must work reliably in dense traffic.
- Optical Flow provides accurate motion information but becomes noisy when many vehicles are present.
- Motion Estimation is more stable in crowded scenes.
- Although it provides less detail, stability is more important for high-density traffic.

Conclusion: Motion Estimation is more suitable because robustness in dense traffic is the priority.

4. Background Modeling Evaluation
A GMM-based system produces:
- Static scenes: 90% accuracy
- Dynamic scenes: 65% accuracy

Requirement:
System must maintain ≥80% accuracy across all scenarios
Compare performance and evaluate whether GMM alone meets requirements. Justify logically and suggest a decision.

ANSWER:

Background Modeling Evaluation

Scene	GMM Accuracy	Requirement	Result
Static	90%	≥80%	✅ Pass
Dynamic	65%	≥80%	❌ Fail

Decision: GMM alone is not sufficient.

- The required accuracy is at least 80% in all scenarios.
- GMM achieves 90% in static scenes, so it passes there.
- In dynamic scenes, accuracy drops to 65%, which is below the required 80%.
- Therefore, GMM alone cannot satisfy the overall requirement.

Suggested decision: Use GMM with an additional technique such as adaptive background modeling or motion-based analysis to improve performance in dynamic scenes.

Conclusion: Do not use GMM alone because it fails the dynamic-scene requirement.

5. Depth Estimation Strategy
Constraints:
- Only one camera available
- Moderate depth accuracy acceptable

Options:
Stereo Vision: high accuracy, needs two cameras
Structure from Motion: moderate accuracy, single camera
Compare both approaches and evaluate which is feasible. Justify using constraint-based reasoning.

ANSWER:

Method	**Accuracy**	**Camera Requirement**	**Feasibility**
Stereo Vision	High	2 cameras	❌ Not feasible
Structure from Motion	Moderate	1 camera	✅ Feasible

Decision: Structure from Motion (SfM)

- Only one camera is available.
- Stereo Vision requires two cameras, so it cannot be used under the given hardware constraint.
- Structure from Motion works with a single camera.
- Its moderate depth accuracy is acceptable because the requirement does not demand high accuracy.

Conclusion: Structure from Motion is the feasible choice because it works with one camera and provides the required moderate depth accuracy.